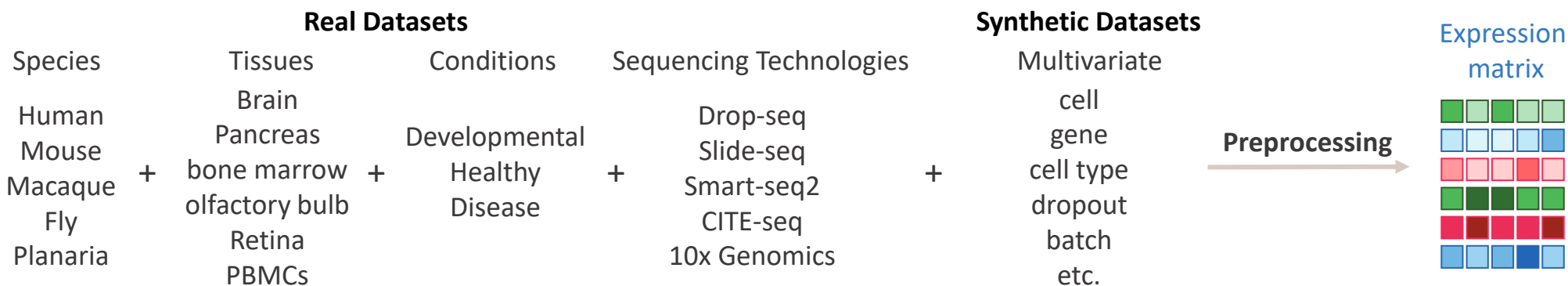


50 real datasets and 50 simulated datasets



26 Dimensionality Reduction Methods

Linear and Probabilistic Factor Models

PCA, SSNMDI
pCMF, tGPLVM
ZIFA, scGBM
GLMPCA

Deep Autoencoders and Generative Models

VAE, scGAE
scvis, SCDRHA
VASC, DREAM
SAUCIE, DRA
scScope

Graph-Based and Diffusion Geometry Models

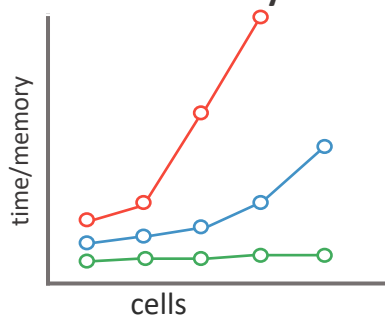
UMAP
PHATE
SIMLR
EDGE
SPDR

Metric Learning and Structure-Aware Embedding

t-SNE
TriMap
PaCMAP
IVIS
SQuAD-MDS

Five evaluation dimensions

Efficiency



Cluster Accuracy

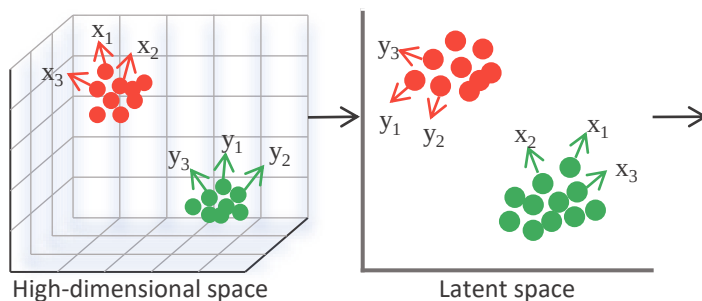


ARI
NMI
SIL
COM
HOM

Stability

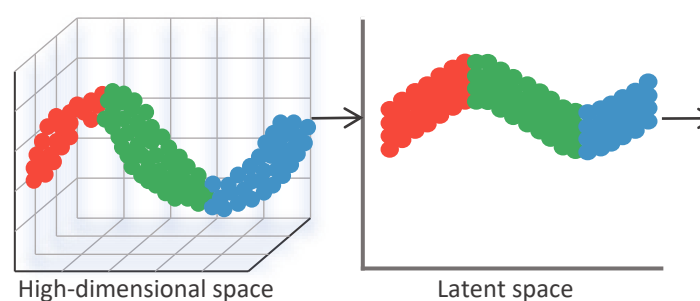
cell	gene	cell type
dropout	batch number	batch strength
outlier prob.	DE prob.	DE strength

Local Structural Preservation



NKR
AJI
...
MRRE
T/C

Global Structural Preservation



Triplet Accuracy
Spearman Cor
...
Pearson Cor
Qglobal